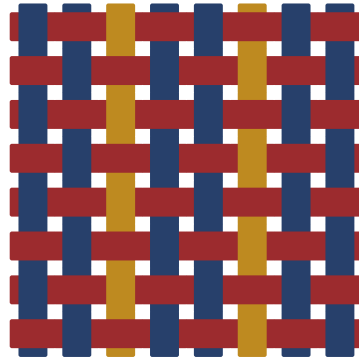




# Information Theory

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Lecture 3 · channel capacity

Oxford B8.4 · 2024--25

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

# THE ONE IDEA, AND THE MAP

□□□□

A channel is nothing but a conditional law  $p(y | x)$ . We cannot change that law, but we may choose how often we use each input. **Capacity** is the best mutual information such a choice can buy:  $C = \max_{p(x)} I(X; Y)$ . Every example in this lecture is one instance of the same two-step move --- split  $I(X; Y) = H(Y) - H(Y | X)$ , then make the output as uncertain as the channel allows.

## The map so far

| stop | object                                                  | question                                        |
|------|---------------------------------------------------------|-------------------------------------------------|
| 1    | $(\mathcal{X}, p(y   x), \mathcal{Y})$                  | What data specify a channel?                    |
| 2    | $p(y^n   x^n) = \prod_i p(y_i   x_i)$                   | What does “memoryless” rule out?                |
| 3    | $C = \max_{p(x)} I(X; Y)$                               | How much information can one channel use carry? |
| 4    | $C = 1$ (noiseless binary)                              | What does a perfect bit pipe look like?         |
| 5    | $H(X   Y) = 0$                                          | When does noise cost nothing at all?            |
| 6    | $C = 1 - h(p)$ (BSC)                                    | What do flipped bits cost?                      |
| 7    | $C = 1 - \alpha$ (BEC)                                  | What do lost bits cost?                         |
| 8    | $C = \log  \mathcal{Y}  - H(r)$                         | What does symmetry buy us for free?             |
| 9    | $0 \leq C \leq \log \min( \mathcal{X} ,  \mathcal{Y} )$ | Which values of $C$ are possible at all?        |

## Conventions

Alphabets  $\mathcal{X}$  and  $\mathcal{Y}$  are finite. All logarithms are base 2, so entropies and capacities are in bits per channel use. We write  $h(p) = -p \log p - (1 - p) \log(1 - p)$  for the binary entropy function, and  $H(r)$  for the entropy of a probability vector  $r$ . The input distribution  $p(x)$  is ours to choose; the transition law  $p(y | x)$  is not.

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## LECTURE 3 · CHANNEL CAPACITY

First fix what a channel is and what memorylessness means. Then define capacity as a maximisation over input distributions, and compute it in the four canonical examples: the noiseless binary channel, a channel whose outputs give the input away, the binary symmetric channel, and the binary erasure channel. Finally isolate the structural reason those computations were easy --- symmetry --- and record the general properties of  $C$ .

### 1 THE CHANNEL AS A CONDITIONAL LAW

□□□□ A channel is a table of conditional probabilities; capacity is the best use we can make of it.

Everything about a discrete channel is contained in one stochastic matrix. The matrix is given to us by physics; what remains free is the distribution with which we feed the inputs. Capacity is the value of that freedom.

**Trigger.** Reach for this model whenever the corruption of a symbol depends only on that symbol.

#### The data of a channel

**Definition 1.1** (*discrete channel*). A **discrete channel** is a system consisting of an input alphabet  $\mathcal{X}$ , an output alphabet  $\mathcal{Y}$ , and a probability transition matrix  $p(y | x)$ , which gives the probability of observing the output symbol  $y$  when the input symbol  $x$  is sent. Being a stochastic matrix means

$$p(y | x) \geq 0 \quad \text{for all } x, y, \quad \sum_{y \in \mathcal{Y}} p(y | x) = \underline{1} \quad \text{for every } x \in \mathcal{X}.$$

The channel is **discrete** because both alphabets are finite.

**Definition 1.2** (*memoryless*). The channel is **memoryless** if the output at each time depends only on the input at that same time, and not on earlier inputs or outputs. Equivalently, for every  $n$  and every  $x^n \in \mathcal{X}^n$ ,  $y^n \in \mathcal{Y}^n$ ,

$$p(y^n | x^n) = \prod_{i=1}^n \underline{p(y_i | x_i)}.$$

A channel that is both discrete and memoryless is called a **discrete memoryless channel** (DMC).

**Remark 1.3.** The product formula quietly assumes two things: that the channel does not remember past inputs (memorylessness) and that the input  $x_i$  is not allowed to depend on the past

outputs  $y^{i-1}$  (no feedback). Both are needed for the factorisation as written, and both hold in the setting of this lecture.

## Capacity

**Definition 1.4** (*information channel capacity*). The (“information”) **channel capacity** of a discrete memoryless channel is

$$C = \max_{p(x)} I(X; Y),$$

where the maximum is taken over all possible input distributions  $p(x)$ . The joint law is always  $p(x, y) = p(x) p(y | x)$ , so choosing  $p(x)$  chooses both the output distribution  $p(y) = \sum_x p(x) p(y | x)$  and the value of  $I(X; Y)$ .

**Remark 1.5.** Capacity also has an **operational** definition: the largest rate, in bits per channel use, at which information can be sent with error probability tending to 0. Shannon’s second theorem (the channel coding theorem) says that the information capacity and the operational capacity coincide — which is why the word “information” is usually dropped. This lecture computes the  $\max_{p(x)} I(X; Y)$  side; the coding theorem is the next lecture’s business.

### The one move used in every example

Because  $I(X; Y) = H(Y) - H(Y | X)$  and

$$H(Y | X) = \sum_x p(x) H(Y | X = x),$$

the second term is an average of *fixed* row entropies of the transition matrix. So maximising  $I(X; Y)$  means making  $H(Y)$  large while keeping the average row entropy small. When all rows have the *same* entropy, the second term does not depend on  $p(x)$  at all and only  $H(Y)$  has to be maximised — this is the shortcut behind Sections 3 and 5.

**Your turn.** Write down the two-term decomposition of  $I(X; Y)$  that this lecture uses, and say which term the channel fixes and which term the input distribution moves.  $I(X; Y) = H(Y) - H(Y | X)$  with  $H(Y | X) = \sum_x p(x) H(Y | X = x)$ . The channel fixes each row entropy  $H(Y | X = x)$ ; the input distribution chooses their weights, and through  $p(y) = \sum_x p(x) p(y | x)$  it also controls  $H(Y)$ .

? What two assumptions are hidden in  $p(y^n | x^n) = \prod_i p(y_i | x_i)$ ?

## ■ 2 TWO CHANNELS WHERE NOISE IS FREE

□□□□ Noise is harmless as long as it never makes two inputs look alike.

Start with the two easy extremes. If the channel is noiseless, one error-free bit per use is obvious. If the channel is noisy but different inputs produce disjoint sets of outputs, the receiver still recovers the input exactly --- so the capacity is again that of a clean pipe. What costs capacity is not randomness, but *confusability*.

**Trigger.** Before computing anything, ask whether the output ever leaves the input in doubt.

### The noiseless binary channel

**Example 2.1** (*noiseless binary channel*). Let  $\mathcal{X} = \mathcal{Y} = \{0, 1\}$  and let every input be reproduced exactly.

0  $\longrightarrow$  0

1  $\longrightarrow$  1

One error-free bit is transmitted per use of the channel, so we expect  $C = 1$ . The definition confirms it: here  $Y = X$ , hence  $H(Y | X) = 0$  and

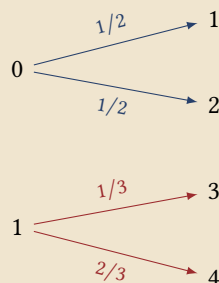
$$C = \max_{p(x)} I(X; Y) = \max_{p(x)} H(X) = \underline{1} \text{ bit,}$$

the maximum being attained at  $p(x) = \underline{\left(\frac{1}{2}, \frac{1}{2}\right)}$ .

### A noisy channel with non-overlapping outputs

**Example 2.2** (*non-overlapping outputs*). Let  $\mathcal{X} = \{0, 1\}$  and  $\mathcal{Y} = \{1, 2, 3, 4\}$ , with

$$p(1 | 0) = p(2 | 0) = \frac{1}{2}, \quad p(3 | 1) = \frac{1}{3}, \quad p(4 | 1) = \frac{2}{3}.$$



The channel is genuinely noisy: the output is random for both inputs. But the output sets  $\{1, 2\}$  and  $\{3, 4\}$  are **disjoint**, so from  $y$  the receiver can always tell which input was sent. Hence

$$H(X | Y) = \underline{0}, \quad I(X; Y) = H(X) - H(X | Y) = H(X),$$

and therefore

$$C = \max_{p(x)} H(X) = 1 \text{ bit},$$

again attained by the uniform input. Randomness inside each row is irrelevant; only overlap between rows costs capacity.

**Remark 2.3.** It is worth naming the general principle: the capacity of a channel depends on its transition matrix only through which inputs can be *confused*. If the supports of distinct rows are pairwise disjoint, the channel is as good as a noiseless channel on  $|\mathcal{X}|$  symbols and  $C = \log |\mathcal{X}|$ .

**Your turn. Noisy typewriter.** Let  $\mathcal{X} = \mathcal{Y} = \{A, B, \dots, Z\}$  (26 letters) and suppose each letter is either printed correctly or printed as the next letter, each with probability  $1/2$ . By using only every other letter of the alphabet as an input, the outputs of the used inputs become disjoint. What capacity does this suggest, and why can it not be beaten? Using 13 letters spaced two apart makes the output sets disjoint, so those 13 inputs form a noiseless channel and  $C \geq \log 13$ . Conversely each row has entropy  $H(Y | X = x) = 1$  bit, so  $I(X; Y) = H(Y) - 1 \leq \log 26 - 1 = \log 13$ . Hence  $C = \log 13$  bits, attained by the uniform distribution on all 26 inputs as well.

? Which feature of the transition matrix actually destroys capacity: randomness within rows, or overlap between rows?

### 3 THE BINARY SYMMETRIC CHANNEL

□□□□ The simplest channel with errors, and already the whole difficulty.

In the binary symmetric channel each input bit is complemented with probability  $p$ . It is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem. Its capacity is  $1 - h(p)$ : one bit per use minus the entropy the noise injects.

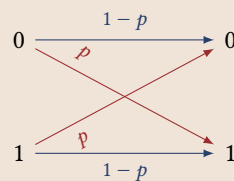
**Trigger.** Use  $I = H(Y) - H(Y | X)$  whenever every row of the transition matrix has the same entropy.

#### The channel

**Definition 3.1** (*binary symmetric channel*). The **binary symmetric channel**  $\text{BSC}(p)$  has  $\mathcal{X} = \mathcal{Y} = \{0, 1\}$  and transition matrix

$$p(y | x) = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix},$$

so the input symbol is complemented with probability  $p$ .



Both rows are the vector  $(1-p, p)$  up to reordering, so every row entropy equals  $\underline{h(p)}$ .

#### Capacity

**Theorem 3.2** (*capacity of the BSC*). The capacity of  $\text{BSC}(p)$  is

$$C = 1 - h(p) \quad \text{bits per channel use,}$$

attained by the uniform input distribution  $p(x) = (\frac{1}{2}, \frac{1}{2})$ .

*Proof.* For any input distribution,

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y | X) \\ &= H(Y) - \sum_x p(x) H(Y | X = x) \\ &= H(Y) - \sum_x p(x) h(p) \\ &= H(Y) - h(p) \\ &\leq 1 - h(p), \end{aligned}$$

where the last step holds because  $Y$  is a binary random variable, so  $H(Y) \leq \log |\mathcal{Y}| = 1$ . Equality requires  $Y$  to be uniform. Taking  $p(x) = (\frac{1}{2}, \frac{1}{2})$  gives

$$\mathbb{P}(Y = 0) = \frac{1}{2}(1 - p) + \frac{1}{2}p = \frac{1}{2},$$

so  $H(Y) = 1$  and the bound is achieved. Hence  $C = 1 - h(p)$ . ■

**Remark 3.3.** The step that makes the computation short is that  $H(Y | X) = h(p)$  *independently of  $p(x)$* : the weights  $p(x)$  multiply identical row entropies. This is exactly the symmetry that Section 5 generalises.

### Reading the answer

| $p$  | $C = 1 - h(p)$ | interpretation                                               |
|------|----------------|--------------------------------------------------------------|
| 0    | 1              | noiseless channel; one clean bit per use                     |
| 1    | 1              | every bit is flipped, which the receiver can simply undo     |
| 1/2  | 0              | the output is independent of the input; nothing gets through |
| 0.11 | $\approx 0.5$  | about one usable bit per two channel uses                    |

~~~~~  
Guessing the uniform input works here only because of symmetry; see Section 6 otherwise.

**Your turn.** Explain in one line why BSC(1) has the same capacity as BSC(0), and why  $p = \frac{1}{2}$  is the worst case. At  $p = 1$  the flip is deterministic, so relabelling the outputs recovers a noiseless channel and  $h(1) = 0$ . At  $p = \frac{1}{2}$  each output is equally likely whatever was sent, so  $Y$  is independent of  $X$ ,  $h(\frac{1}{2}) = 1$  and  $C = 0$ .

**Your turn.** Reconstruct the chain of five lines in the proof, naming the reason for each step.  $I = H(Y) - H(Y | X)$  (definition); expand  $H(Y | X) = \sum_x p(x)H(Y | X = x)$  (definition of conditional entropy); each  $H(Y | X = x) = h(p)$  (rows of the matrix); so the sum is  $h(p)$  since  $\sum_x p(x) = 1$ ; finally  $H(Y) \leq 1$  because  $Y$  is binary, with equality iff  $Y$  is uniform, which the uniform input achieves.

? Why is  $H(Y | X)$  constant in  $p(x)$  for the BSC?

## 4 THE BINARY ERASURE CHANNEL

□□□□ Losing a bit is cheaper than being lied to about it.

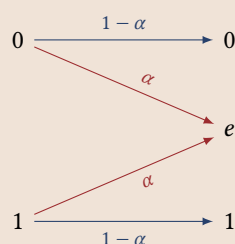
The erasure channel loses a fraction  $\alpha$  of the bits but never corrupts one: the receiver always knows *which* bits were lost. Its capacity is  $1 - \alpha$  --- exactly the fraction of channel uses that survive. Compare this with the BSC, where a flip probability of  $\alpha$  costs  $h(\alpha)$ , which is much more.

**Trigger.** When one output symbol means “no information”, condition on the event that it occurred.

### The channel

**Definition 4.1** (*binary erasure channel*). The **binary erasure channel**  $\text{BEC}(\alpha)$  has  $\mathcal{X} = \{0, 1\}$  and  $\mathcal{Y} = \{0, 1, e\}$ , where  $e$  is the **erasure** symbol, and

$$p(0 | 0) = p(1 | 1) = 1 - \alpha, \quad p(e | 0) = p(e | 1) = \alpha.$$



No bit is ever complemented, so an output in  $\{0, 1\}$  identifies the input with certainty; an output  $e$  says nothing at all. Every row is a permutation of  $(1 - \alpha, \alpha, 0)$ , hence  $H(Y | X) = \underline{h(\alpha)}$ .

### Capacity

**Theorem 4.2** (*capacity of the BEC*). The capacity of  $\text{BEC}(\alpha)$  is

$$C = 1 - \alpha \quad \text{bits per channel use,}$$

attained by the uniform input distribution.

*Proof.* Write, as always,

$$C = \max_{p(x)} I(X; Y) = \max_{p(x)} [H(Y) - H(Y | X)] = \max_{p(x)} H(Y) - h(\alpha).$$

To compute  $H(Y)$ , let  $E = \mathbf{1}\{Y = e\}$  be the indicator of an erasure. Since  $E$  is a function of  $Y$ , we have  $H(Y, E) = H(Y)$ , and the chain rule gives

$$H(Y) = H(Y, E) = H(E) + H(Y | E).$$

Now  $E \sim \text{Bernoulli}(\alpha)$ , so  $H(E) = h(\alpha)$ . Given  $E = 1$  the output is  $e$  with certainty, contributing nothing; given  $E = 0$  the output equals the input. Writing  $\mathbb{P}(X = 1) = \pi$ ,

$$H(Y | E) = \alpha \cdot 0 + (1 - \alpha)h(\pi), \quad \text{so} \quad H(Y) = h(\alpha) + (1 - \alpha)h(\pi).$$

Therefore

$$C = \max_{\pi} [h(\alpha) + (1 - \alpha)h(\pi)] - h(\alpha) = \max_{\pi} (1 - \alpha)h(\pi) = 1 - \alpha,$$

the capacity being achieved at  $\pi = \frac{1}{2}$ . ■

**Remark 4.3.** The interpretation is exact rather than approximate: a fraction  $\alpha$  of the transmitted bits is lost, so at most a fraction  $1 - \alpha$  can get through, and that much is achievable. Note also that here the two “obvious” bounds behave differently:  $H(Y) \leq \log 3$  is *not* tight, because the erasure symbol can never be reached more often than  $\alpha$  of the time. Splitting  $H(Y)$  via the erasure indicator is what replaces the crude bound.

### BSC versus BEC at the same noise level

| channel         | capacity     | what the receiver knows                                            |
|-----------------|--------------|--------------------------------------------------------------------|
| BSC( $p$ )      | $1 - h(p)$   | every received bit may be a lie; the location of errors is unknown |
| BEC( $\alpha$ ) | $1 - \alpha$ | received bits are certainly correct; erasures are flagged          |

At  $p = \alpha = 0.1$  these give  $C \approx 0.531$  and  $C = 0.9$ : knowing *where* the damage is worth a lot.

**Your turn.** The key trick was  $H(Y) = H(E) + H(Y | E)$ . State the two facts that make this identity legitimate and useful here. First,  $E = \mathbf{1}\{Y = e\}$  is a deterministic function of  $Y$ , so  $H(Y, E) = H(Y)$  and the chain rule may be applied to the pair. Second, conditioning on  $E$  splits the output into a degenerate case ( $Y = e$ , entropy 0) and a noiseless case ( $Y = X$ , entropy  $h(\pi)$ ), which is why  $H(Y | E) = (1 - \alpha)h(\pi)$ .

**Your turn.** Complete the two capacities from memory: BSC( $p$ ) has  $C = \underline{1 - h(p)}$  and BEC( $\alpha$ ) has  $C = \underline{1 - \alpha}$ .

? Why is  $H(Y) \leq \log 3$  useless for the BEC?

## 5 SYMMETRIC CHANNELS

□□□□ When the rows all look alike, the uniform input is optimal and capacity is a one-line formula.

The BSC computation used only two facts: all rows of the transition matrix have the same entropy, and the uniform input produces the uniform output. Channels with those two features are called (weakly) symmetric, and for them capacity is  $\log|\mathcal{Y}| - H(r)$  with  $r$  any row --- no optimisation needed.

**Trigger.** Check symmetry first; if the channel is symmetric, capacity costs one line instead of a maximisation.

### Definitions

**Definition 5.1** (*symmetric channel*). A discrete memoryless channel is **symmetric** if the rows of its transition matrix  $p(y | x)$  are permutations of each other, and the columns are permutations of each other as well.

**Definition 5.2** (*weakly symmetric channel*). A channel is **weakly symmetric** if the rows of  $p(y | x)$  are permutations of each other and every column has the same sum  $\sum_x p(y | x)$ . Every symmetric channel is weakly symmetric, since permuted columns have equal sums.

**Example 5.3** (*a symmetric channel*). Let  $\mathcal{X} = \mathcal{Y} = \{1, 2, 3\}$  with

$$p(y | x) = \begin{pmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}.$$

Each row is a permutation of  $r = (0.3, 0.2, 0.5)$ , and each column is a permutation of the same vector, so the channel is symmetric.

### Capacity of a weakly symmetric channel

**Theorem 5.4** (*capacity by symmetry*). For a weakly symmetric channel with row vector  $r$  (any row of the transition matrix),

$$C = \log|\mathcal{Y}| - H(r),$$

and the capacity is achieved by the uniform distribution on  $\mathcal{X}$ .

*Proof.* Since all rows are permutations of  $r$ , every row entropy equals  $H(r)$ , so for any input

distribution

$$I(X; Y) = H(Y) - H(Y | X) = H(Y) - \sum_x p(x) H(r) = H(Y) - H(r) \leq \log |\mathcal{Y}| - H(r),$$

with equality if and only if  $Y$  is uniform on  $\mathcal{Y}$ . It remains to show that the uniform input makes  $Y$  uniform. Let  $c = \sum_x p(y | x)$  be the common column sum. With  $p(x) = 1/|\mathcal{X}|$ ,

$$p(y) = \sum_x \frac{1}{|\mathcal{X}|} p(y | x) = \frac{c}{|\mathcal{X}|},$$

which does not depend on  $y$ ; being a probability distribution it must equal  $1/|\mathcal{Y}|$ . Hence the bound is attained and  $C = \log |\mathcal{Y}| - H(r)$ . ■

**Example 5.5** (capacity of Example 5.3). With  $r = (0.3, 0.2, 0.5)$ ,

$$H(r) = -0.3 \log 0.3 - 0.2 \log 0.2 - 0.5 \log 0.5 \approx 1.485, \quad \log |\mathcal{Y}| = \log 3 \approx 1.585,$$

so

$$C = \log 3 - H(r) \approx \underline{0.0995} \text{ bits},$$

attained by the uniform input  $p(x) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ . The channel is close to useless: its rows are nearly uniform, so the output barely depends on the input.

### Who is symmetric and who is not

| channel          | symmetric? | capacity via the formula                                                     |
|------------------|------------|------------------------------------------------------------------------------|
| noiseless binary | yes        | $\log 2 - H(1, 0) = 1$                                                       |
| BSC( $p$ )       | yes        | $\log 2 - h(p) = 1 - h(p)$                                                   |
| Example 5.3      | yes        | $\log 3 - H(r) \approx 0.0995$                                               |
| BEC( $\alpha$ )  | no         | column sums $1 - \alpha, 1 - \alpha, 2\alpha$ differ; formula does not apply |

The BEC row vectors *are* permutations of one another, so the row condition alone is not enough: without equal column sums the uniform input need not give a uniform output, and indeed the true answer  $1 - \alpha$  differs from  $\log 3 - h(\alpha)$ .

**Your turn.** State the two places where symmetry is used in the proof, one for each of the two defining conditions. Rows being permutations of each other makes every  $H(Y | X = x)$  equal to  $H(r)$ , so  $H(Y | X) = H(r)$  regardless of  $p(x)$ . Equal column sums make  $p(y)$  constant under the uniform input, so  $H(Y) = \log |\mathcal{Y}|$  is attained and the inequality becomes an equality.

**Your turn.** Compute the capacity of the channel with  $\mathcal{X} = \mathcal{Y} = \{1, 2, 3\}$  and every row a permutation of  $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$  arranged so that columns are permutations too.  $H(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 2 = 1.5$  bits, so  $C = \log 3 - 1.5 \approx 1.585 - 1.5 = 0.085$  bits.

? Which condition fails for the BEC, and what goes wrong if you apply the formula anyway?

## 6 PROPERTIES OF CAPACITY

□□□□ A concave maximisation over a simplex: the maximum exists, and there is a test for it.

Once the easy examples are done, two questions remain. Which values can  $C$  take? And what to do when no symmetry hands us the optimal input? The answers:  $C$  lies between 0 and  $\log \min(|\mathcal{X}|, |\mathcal{Y}|)$ , and  $I(X; Y)$  is a concave function of  $p(x)$ , so the maximisation is a well-behaved convex problem with a verifiable optimality condition.

**Trigger.** Use the equal-divergence test when you have guessed a candidate optimal input and want to prove it optimal.

### Range

**Proposition 6.1** (*elementary bounds*). For every discrete memoryless channel,

$$0 \leq C \leq \log \min(|\mathcal{X}|, |\mathcal{Y}|).$$

Moreover  $C = 0$  if and only if  $X$  and  $Y$  are independent for every input distribution, i.e. all rows of  $p(y | x)$  are equal.

*Proof.* Non-negativity holds because  $I(X; Y) \geq 0$  always, so the maximum is at least 0. For the upper bound,

$$I(X; Y) \leq H(X) \leq \log |\mathcal{X}|, \quad I(X; Y) \leq H(Y) \leq \log |\mathcal{Y}|,$$

using  $I(X; Y) = H(X) - H(X | Y) \leq H(X)$  and the symmetric identity. Taking the smaller of the two bounds gives the claim. If all rows are equal then  $p(y) = p(y | x)$  for every  $x$ , so  $Y$  is independent of  $X$  and every mutual information vanishes; conversely  $C = 0$  forces  $I(X; Y) = 0$  for the uniform input, whose support is all of  $\mathcal{X}$ , which forces  $p(y | x) = p(y)$  for all  $x$ . ■

**Your turn.** Which channel of this lecture attains each of the two extremes  $C = 0$  and  $C = \log \min(|\mathcal{X}|, |\mathcal{Y}|)$ ?  $C = 0$  for BSC( $\frac{1}{2}$ ), whose two rows are both  $(\frac{1}{2}, \frac{1}{2})$ . The upper extreme  $C = \log 2 = 1$  is attained by the noiseless binary channel and also by the channel with non-overlapping outputs, where  $|\mathcal{X}| = 2$  and  $H(X | Y) = 0$ .

### Shape of the optimisation

**Proposition 6.2** (*concavity in the input*). For a fixed transition law  $p(y | x)$ , the map  $p(x) \mapsto I(X; Y)$  is continuous and **concave** on the probability simplex over  $\mathcal{X}$ .

*Proof.* Write  $I(X; Y) = H(Y) - H(Y | X)$ . The second term is  $\sum_x p(x) H(Y | X = x)$ , which is linear in  $p(x)$ . The first term is  $H$  evaluated at  $p(y) = \sum_x p(x) p(y | x)$ , which is a linear map of  $p(x)$ ; since  $H$  is concave and the composition of a concave function with a linear map is concave,  $p(x) \mapsto H(Y)$  is concave. A concave function minus a linear function is concave. Continuity is clear from the formulas. ■

The consequences are the practical ones: the simplex is compact, so the maximum is attained; every local maximum is global; and standard convex-optimisation tools (or the Blahut–Arimoto algorithm) compute  $C$  numerically.

**Remark 6.3.** Note the contrast with source coding, where the relevant convexity was in the distribution being coded. Here the distribution is the *design variable* and the channel is fixed. Mutual information is concave in  $p(x)$  for fixed  $p(y | x)$ , and convex in  $p(y | x)$  for fixed  $p(x)$ .

## A test for optimality

**Theorem 6.4** (*equal-divergence condition*). An input distribution  $p^*(x)$  with induced output distribution  $p^*(y)$  achieves capacity if and only if there is a constant  $C$  with

$$D(p(y | x) \| p^*(y)) \begin{cases} = C, & \text{for every } x \text{ with } p^*(x) > 0, \\ \leq C, & \text{for every } x \text{ with } p^*(x) = 0, \end{cases}$$

and in that case the constant equals the capacity.

The identity behind it is  $I(X; Y) = \sum_x p(x) D(p(y | x) \| p(y))$ : the mutual information is the average divergence between a row of the channel and the output distribution. If two used inputs had different divergences, shifting weight towards the larger one would increase the average — so at the optimum all used rows must be equidistant from  $p^*(y)$ .

**Example 6.5** (*checking the BEC answer*). For  $\text{BEC}(\alpha)$  the uniform input gives  $p^*(y) = (\frac{1-\alpha}{2}, \frac{1-\alpha}{2}, \alpha)$  on  $\{0, 1, e\}$ . For the input  $x = 0$ , whose row is  $(1 - \alpha, 0, \alpha)$ ,

$$D(p(y | 0) \| p^*(y)) = (1 - \alpha) \log \frac{1 - \alpha}{(1 - \alpha)/2} + \alpha \log \frac{\alpha}{\alpha} = 1 - \alpha,$$

and by symmetry the same holds for  $x = 1$ . Both used inputs have divergence  $1 - \alpha$ , which confirms both the optimality of the uniform input and the value  $C = 1 - \alpha$ .

**Your turn.** Carry out the same check for  $\text{BSC}(p)$ : with the uniform input the output is uniform, so compute  $D((1 - p, p) \| (\frac{1}{2}, \frac{1}{2}))$  and compare with  $1 - h(p)$ .  $D((1 - p, p) \| (\frac{1}{2}, \frac{1}{2})) = (1 - p) \log \frac{1-p}{1/2} + p \log \frac{p}{1/2} = 1 + (1 - p) \log(1 - p) + p \log p = 1 - h(p)$ , identical for the row of  $x = 1$ . So the uniform input is optimal and  $C = 1 - h(p)$ , as computed directly.

## Summary card

| channel                       | $C$                         | optimal input                               |
|-------------------------------|-----------------------------|---------------------------------------------|
| noiseless binary              | 1                           | uniform                                     |
| non-overlapping outputs       | $\log  \mathcal{X} $        | uniform                                     |
| noisy typewriter (26 letters) | $\log 13$                   | uniform                                     |
| $\text{BSC}(p)$               | $1 - h(p)$                  | uniform                                     |
| $\text{BEC}(\alpha)$          | $1 - \alpha$                | uniform                                     |
| weakly symmetric, row $r$     | $\log  \mathcal{Y}  - H(r)$ | uniform                                     |
| general                       | $\max_{p(x)} I(X; Y)$       | concave maximisation; equal-divergence test |

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Asymmetric channels, e.g. the Z-channel where only one input can be corrupted, have non-uniform optimal inputs --- a good exercise for this test.

The uniform input is optimal in every line above because each of those channels is symmetric in some sense; asymmetry is what makes the general problem hard.

? What identity turns  $I(X; Y)$  into an average of divergences, and what does it say at the optimum?